

10-23-25 Annotation Plan

Goal: 70+% ML tracking in 2 weeks

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The project goal

Goal: To classify/locate objects better than Tocaro Blue – via any method needed

Sub-Goal: Build a comprehensive dataset with consistent tracking IDs across frames

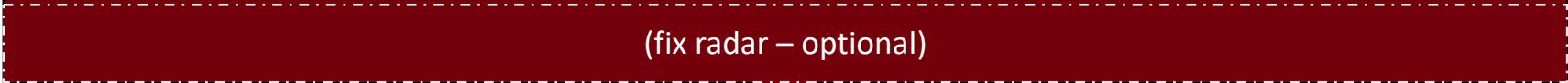
NOT a goal: Develop a novel or interesting ML model.

The rough outline

Build annotation pipeline

A large, solid dark red arrow pointing downwards, connecting the first step to the second.

Collect new data in correct format

A dashed white rectangular box surrounding the text "(fix radar – optional)".

(fix radar – optional)

A large, solid dark red arrow pointing downwards, connecting the second step to the third.

Train a model

Annotation Pipeline

Use existing tracking software to get object locations across frames.



Use object locations to cue SAM



Show SAM output to annotator to verify.

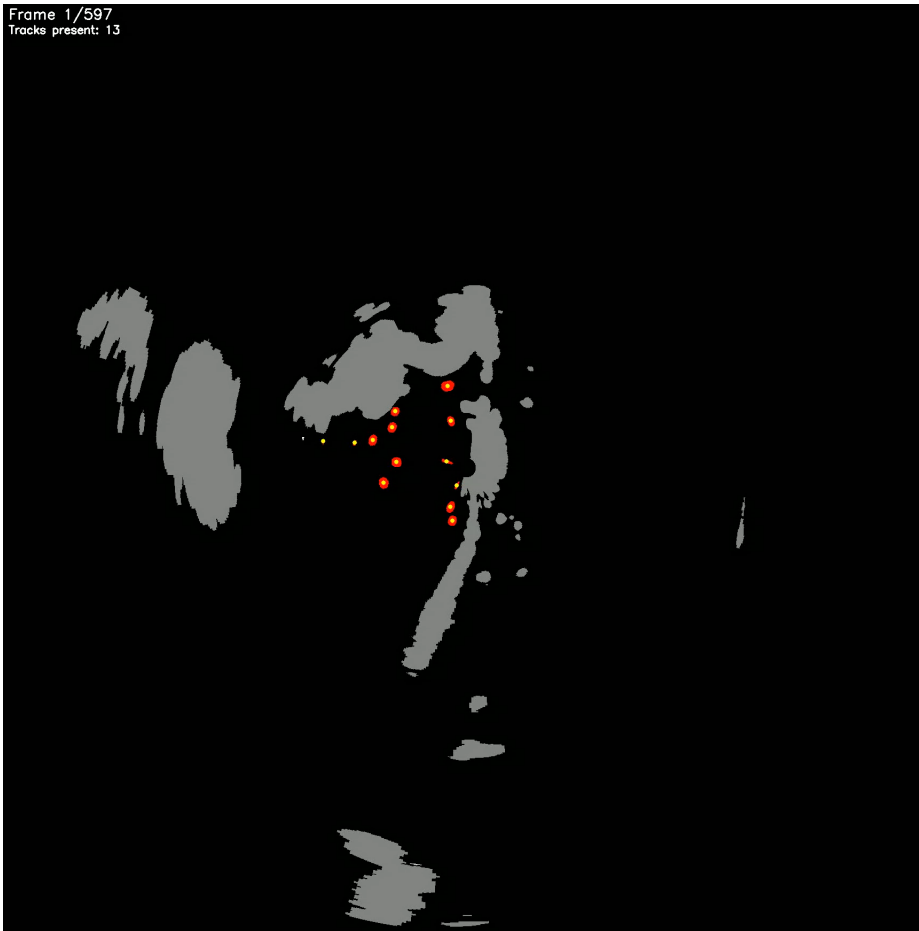


Use final segmentation output later for center points, bounding boxes, etc.

This'll have to be custom or a heavily modified version of Label Studio

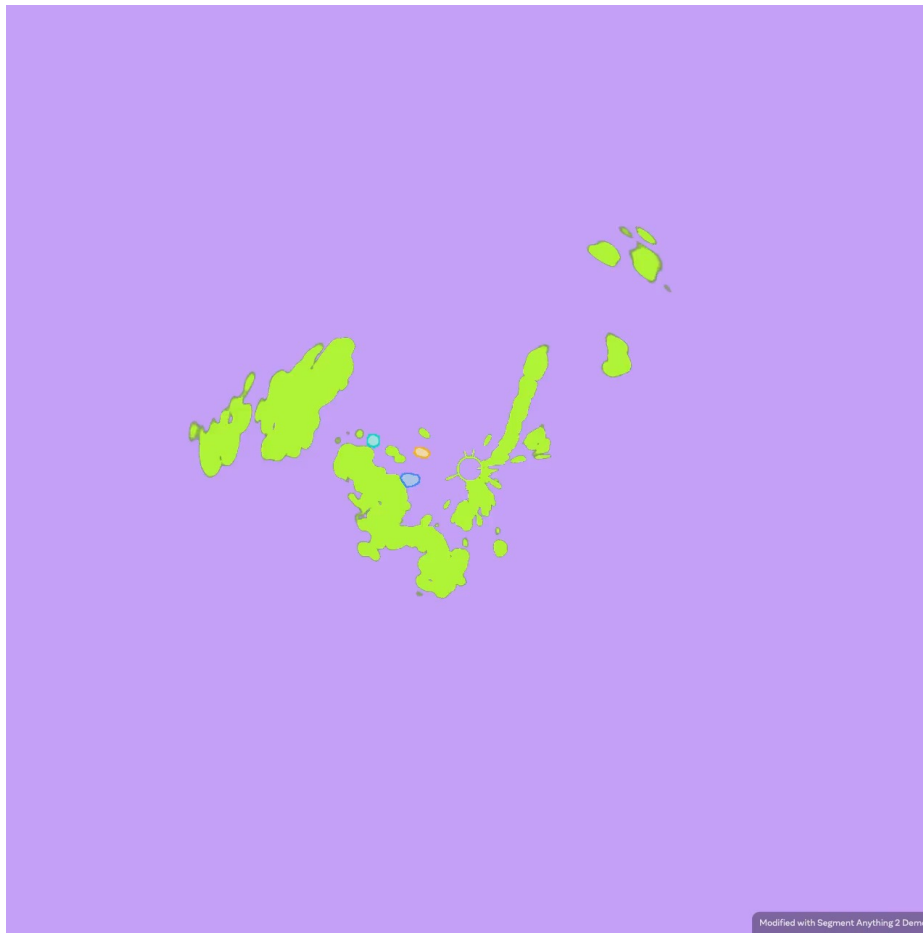
Annotation Pipeline

Frame 1/597
Tracks present: 13

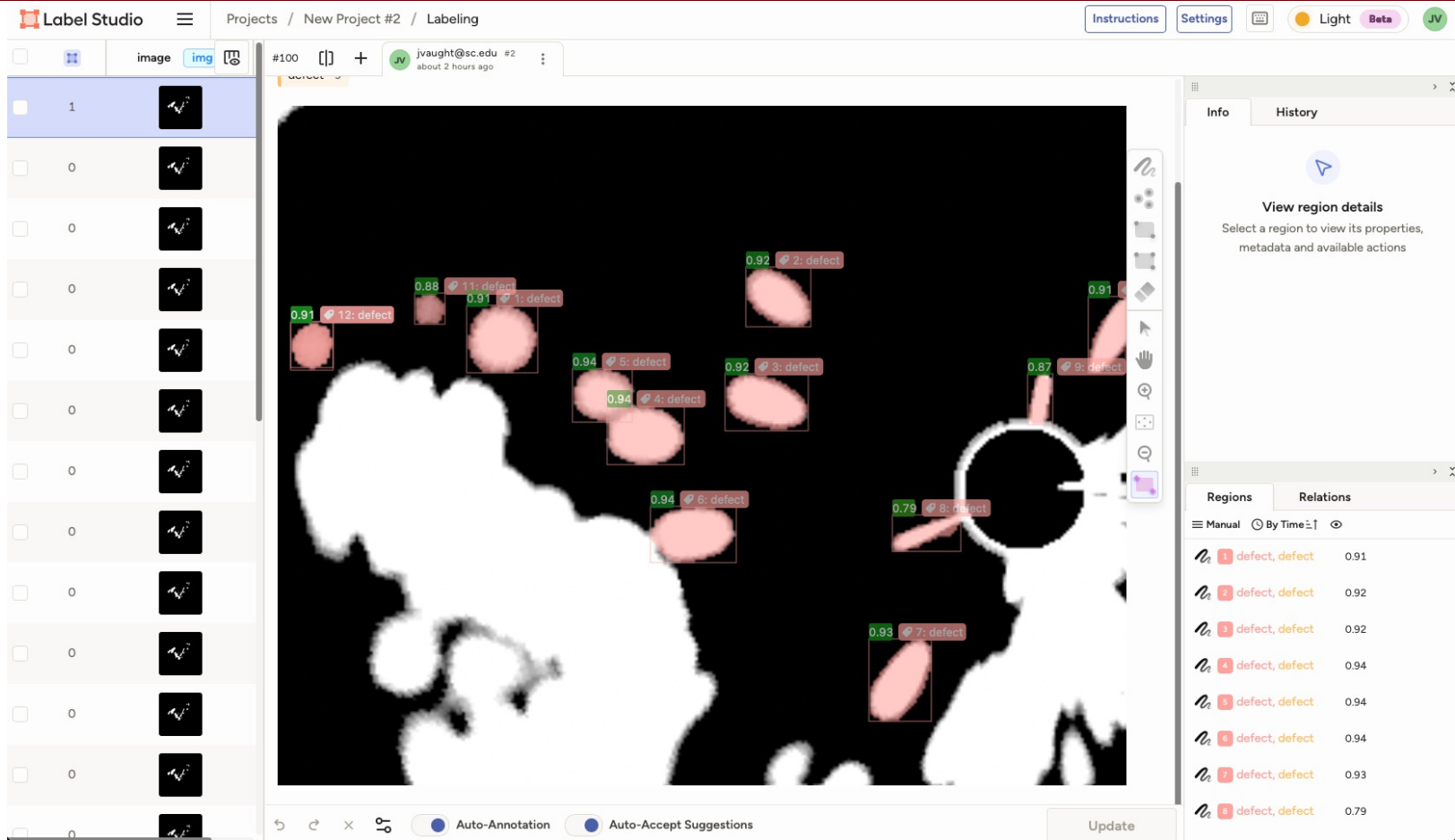


Annotation Pipeline

SAM 2 Demo
annotation



Label Studio + SAM



Data Collection

Charleston on Sunday

Hampton Roads Tue-Thurs

- Going to test Gain ramp – 60,70,80,90,95,100
- Going to test plain longer and shorter pulses
- Then test ramped pulses.
- Change radar settings to use 60 RPM (data loss ok)

Data Collection



RAW Data



Processed Data

Train a Model

GPS + Heading -> fed into land discriminator from internet

Blob/Object detection done on remaining area using YOLO or similar CNN. 70+% accurate

- This will use delay-decay imaging, like most existing algorithms.

Temporal tracker applied to all detected objects – I'll find something for this. 50+% ID tracking.

- No velocity or heading direction to be extracted

Questions?

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